Finding Order and Scale Mistakes on Graphs
Answer Key

Grades 4–5 Mathematics

Quick Answers

1. (a) x-coordinate of B = 5, (a) y-coordinate of B = 2, —
2. 35
3. (a) books on Wednesday = 10, —
4. 3 units
5. 105
6. (a) marks tall on the new axis = 10, —
7. (a) mean number of books per day = 21, (b) range of the daily totals = 25
8. (a) marks up the y-axis = 6, (b) y-value the student's mark really names = 150, —

What is verified

4 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 3, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics
5.G.A.1–A.2 – *problems 1, 4, 8*
5.MD – *problems 2, 3, 5, 6, 7*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

2. *If they got 7: counted each mark on the axis as one book*
3. *If they got 2: counted each mark on the axis as one book*
4. *If they got 0: assumed a reversed pair names the same place, so answered no gap at all*
5. *If they got 21: added the mark counts instead of the book values*
6. *If they got 4: kept using the old interval of 5 books per mark*
7. *If they got 4.2: averaged the mark counts instead of the book values*
8. *If they got 30: used the x-axis interval of 1 on the y-axis as well*

1. Ravi was asked to plot $(2, 5)$ on the Grid. His mark landed on point B. (a) Write B's ordered pair. (b) Explain the mistake.

Solution (a): Read B the way any point is read — across first, then up. It sits 5 squares across and 2 squares up. $\boxed{(5, 2)}$

(b) — instructor-judged. Look for a response naming the order: the first number of an ordered pair counts *across* and the second counts *up*, so Ravi used 5 as the across number when it was meant to be the up number. $(2, 5)$ and $(5, 2)$ are different places built from the same two digits.

Saying only "he put it in the wrong spot" describes the symptom, not the error.

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2. Leo says Thursday shows 7 books because the bar reaches the 7th mark. How many books does Thursday actually show?

Solution: The vertical axis is marked every 5 books, so each mark is worth 5, not 1. Seven marks means 7×5. $\boxed{35}$

Common wrong answer: 7 — the number of marks, read as if the axis counted single books.

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3. Sam counted the marks on the Wednesday bar and reported 2 books. (a) How many books does Wednesday actually show? (b) Explain Sam's mistake.

Solution (a): Two marks, each worth 5 books: 2×5. $\boxed{10}$

(b) — instructor-judged. Look for a response that names the size of one interval: on this axis the gap between marks is 5 books, so counting marks gives the number of *marks*, and each one still has to be multiplied by 5. The first thing to read on any graph is what one interval is worth.

Common wrong answer: 2 — the mark count reported as the book count.

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4. Points C and D are a reversed pair. Measured left to right, how far apart are they?

Solution: C is at $(4, 7)$ and D is at $(7, 4)$. Left-to-right distance uses only the across numbers: $7 - 4$. $\boxed{3 \text{ units}}$

Common wrong answer: 0 — from assuming that a reversed pair names the same place. It does not, and this problem measures exactly how far apart the two readings put you.

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5. Mia added the mark counts and reported 21 books for the week. How many books were actually read?

Solution: Read each bar in books, then add: $15 + 25 + 10 + 35 + 20$. $\boxed{105}$

Common wrong answer: 21 — the sum of the mark counts $3 + 5 + 2 + 7 + 4$. Notice that $21 \times 5 = 105$: her total was right in *marks* and simply never converted to books.

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6. The data is redrawn with the vertical axis marked every 2 books. (a) How many marks tall is the Friday bar now? (b) Were more books read on Friday?

Solution (a): Friday is 20 books. On an axis where one mark is worth 2 books, that takes $20 \div 2$ marks. $\boxed{10}$

(b) — instructor-judged. No. Look for a response saying the data did not change — the same 20 books simply need more marks when each mark is worth less, so the bar is drawn taller. Changing the interval changes the picture, never the amount. A response that compares only the heights has fallen for the trap the question sets.

Common wrong answer: 4 — from keeping the old interval of 5 books per mark on an axis that now uses 2.

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7. A student averaged the mark counts and reported 4.2 books per day. (a) Actual mean. (b) Range of the five daily totals.

Solution (a): Use the book values: $15 + 25 + 10 + 35 + 20 = 105$, and $105 \div 5$. $\boxed{21}$

(b) Largest daily total minus smallest: $35 - 10$. $\boxed{25}$

Common wrong answer: 4.2 — the mean of the mark counts. Again it is out by exactly the interval: $4.2 \times 5 = 21$.

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8. A student plots $(3, 30)$ where the horizontal axis is marked every 1 and the vertical axis every 5. The student counts 3 across and 30 up. (a) How many marks up should they have counted? (b) Counting 30 marks up, what y-value did the mark reach? (c) Explain the two things that went wrong.

Solution (a): Each mark up is worth 5, so reaching 30 takes $30 \div 5$ marks. $\boxed{6}$

(b) Counting 30 marks when each is worth 5 lands at 30×5. $\boxed{150}$

(c) — instructor-judged. Two separate faults, and a full-credit answer names both.

First, the two axes do not share an interval: 3 marks across really is 3, but the same mark-counting habit is wrong going up. Second, having used the number 30 as a mark count instead of a value, the mark ended at 150 — five times too high — so the point plotted is nowhere near $(3, 30)$.

Common wrong answer: 30 in part (a) — from applying the horizontal interval of 1 to the vertical axis.